

Program : Diploma in Chemical Engineering	
Course Code : 4074	Course Title: Sustainable Materials and
Semester : 4	Credits: No Credits
Course Category: Program Core	
Periods per week: 3 (L: 1 T: 1 P: 1)	Periods per semester: 45

Course Objectives:

- To practice the production and handling of sustainable materials

Course Pre-requisites:

Topic	Course code	Course name	Semester
Environmental Science		Environmental Science	2

Course Outcomes :

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Summarize sustainable materials	12	Applying
CO2	Practice the synthesis of bio fertilizers and bio pesticides	12	Applying
CO3	Outline sustainable waste management.	12	Understanding
CO4	Summarise the importance of eco friendly startups	9	Applying

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1					3		
CO2					3		
CO3					3		
CO4						3	3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	Summarize sustainable materials		
M1.01	Summarize the concept of sustainability.	3	Understanding
M1.02	Identify importance of bioplastics	3	Understanding
M1.03	Illustrate manufacturing of bioplastics.	3	Understanding
M1.04	Synthesize bioplastic	3	Applying
Contents: Define sustainability, need, concept, challenges - industrial symbiosis. Define bioplastics- identify the sources of bioplastics - properties of bioplastic- advantages of bioplastics - uses of bioplastics - Growth of bioplastic industry. Illustrate the industrial production of polyhydroxyalkanoates (PHA), uses, comparison with existing plastics. Illustrate the industrial production of poly lactic acid (PLA), uses, comparison with existing plastics. Synthesize bio plastic from starch based sources (lab activity)			
CO2	Practice the synthesis of bio fertilizers and bio pesticides		
M2.01	Outline bio fertilizers.	3	Understanding
M2.02	Outline bio pesticides.	3	Understanding
M2.03	Synthesize bio fertilizer (laboratory activity)	3	Applying
M2.04	Synthesize bio pesticides (laboratory activity)	3	Applying
	Series Test – I		
Contents: Define bio fertilizer, need for bio fertilizer – Classification - Nitrogen fixing bio fertilizers, Phosphate bio fertilizers, Potassium bio fertilizers – Commonly used bio fertilizers.			

Define bio pesticide - need and advantages of bio pesticides in organic farming – Classification, microbial pesticides, biochemical or herbal pesticides - Commonly used bio pesticides.

Synthesis of bio fertilizer (laboratory activity).

Synthesis of bio pesticides (laboratory activity).

CO3	Outline sustainable waste management		
M3.01	Identify 3R principles.	3	Understanding
M3.02	Summarise plastic waste management	3	Understanding
M3.03	Summarise paper waste management	3	Understanding
M3.04	Summarise metallic waste management	3	Understanding

Contents:

The 3 R's of the environment, reduce, reuse, recycle.

Plastic waste management - separation, application of plastic waste - recycling methods - recycling of PVC.

Paper waste management - Manufacturing of recycled paper from paper waste, cellulose recovery.

Metallic waste management – Steel waste management – Aluminum waste management.

CO4	Summarise the importance of eco friendly startups		
M4.01	Summarise the process and concepts of entrepreneurship	1	Understanding
M4.02	Identify the needs for eco friendly startups	2	Understanding
M4.03	Identify an eco friendly startup	2	Understanding
M4.04	Prepare a business model for an eco friendly startup	4	Applying
	Series Test – II		

Contents:

Self-employment – Entrepreneurship - Startup - List the self employment programs in India – TBI - MSME

Needs for eco friendly startups

Case study for an eco friendly startup in your locality [Group activity]

Develop a business plan for an eco friendly startup [Group activity]

Text / Reference

T/R	Book Title/Author
T1	Biofertilizer and Organic Farming; NIIR Board
R1	Bio-based plastics Materials and Applications ; Stephan Kabasci
R2	Handbook of recycling; Ernst Worrel and Markus A Reuter
R3	Entrepreneurship; Dr.Sucheta Gauba
R4	Before you Start up; Pankaj Goyal

Online Resources

Sl.No	Website Link
1	www.shomusbiology.com
2	www.startupindia.gov